

Introduction to Contract Theory and Auctions

Lecture 2

Background Material: Binary Relations & Preference Relations

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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- Given two sets, X and Y , the Cartesian product $X \times Y$ is the set of ordered pairs of elements of X and Y .
- A **relation** between members of X and Y can be thought as a subset of $X \times Y$.
- A **binary relation** on X is a relation over $X \times X$.
- For a binary relation R on X , we have that $R \subseteq X \times X$.
- We usually write $x R y$ rather than $(x, y) \in R$. Read x *is R-related to* y .

A bestiary of binary relations

Each of the following definitions should be interpreted with the proper universal quantifier: “for every x, y, z ” etc. The symbol $\neg$ indicates negation, and a compound expression $x R y$ and $y R z$ may be abbreviated as $x R y R z$.

A binary relation R on a set X is:

reflexive if $x R x$.

irreflexive if $\neg(x R x)$.

symmetric if $x R y$ implies $y R x$. Note that this does not imply reflexivity.

asymmetric if $x R y$ implies $\neg(y R x)$. An asymmetric relation is irreflexive.

antisymmetric if $x R y$ with $x \neq y$ implies $\neg(y R x)$. An antisymmetric relation may or may not be reflexive.

transitive if $x R y$ and $y R z$ imply $x R z$.

complete if either $x R y$ or $y R x$ or both. A complete relation is reflexive.

an equivalence relation if it is reflexive, symmetric and transitive.

the symmetric part of the relation S if $x R y \Leftrightarrow (x S y \& y S x)$.

the asymmetric part of the relation S if $x R y \Leftrightarrow (x S y \& \neg y S x)$.

Definition 1.1 (Preference relation)

A preference relation on the set X is a complete and transitive binary relation on X .

We will model the DM's tastes via preference relations. The idea is as follows: imagine that we present to the DM a questionnaire with the following format

$R(x, y)$ (for all $x, y \in X$, not necessarily distinct): is x **at least as preferred as** y ? Check one and only one of the following two boxes.

☐ Yes

☐ No

In order to be a legal response, we require that the DM checks exactly only one of the boxes in each question.

- To qualify the answers as a preference relation, we need that
 - 1 The answer to at least one of the question $R(x, y)$ and $R(y, x)$ be Yes. In particular, the silly question $R(x, x)$ must always be Yes.
 - 2 For every x, y, z , if the answer to $R(x, y)$ is Yes and the answer to $R(y, z)$ is also Yes, then so is the answer to $R(x, z)$.
- Note that under these conditions, R indeed is a complete and transitive, and thus a preference relation.

Notation and Utility functions

- We use a special symbol to denote the answers to each question $R(x, y)$ from the questionnaire: $x \succsim y$, which you can read as x is **at least as preferred as** y .
- The symmetric part of $\succsim$ is denoted by $\sim$, if $x \sim y$ we read x is **indifferent to** y .
- The asymmetric part of $\succsim$ is denoted by $\succ$, if $x \succ y$ we read x is **strictly preferred to** y .
- You are probably not used to describe the DM's preferences via the primitive of a preference relation. Most likely, you know how to do this via a **utility function**.
- It turns out that, taking as primitive the preference relation, if there exist some utility function that represents them, we say that they the utility function **represents** the preference relation.

Definition 1.2 (Utility function)

For any set X , and **preference relation** $\succsim$ on X , the function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if

$$x \succsim y \text{ if and only if } u(x) \geq u(y).$$

We say that u is a **utility function** for $\succsim$.

When can a preference relation be represented by a utility function?

Definition 1.3 (Minimal and Maximal Alternatives)

For any set X and preference relation $\succsim$ on X , the alternative $x \in X$ is **minimal with respect to $\succsim$ in X** if $y \succsim x$ for all $y \in X$ and is **maximal with respect to $\succsim$ in X** if $x \succsim y$ for all $y \in X$.

Lemma 1.1 (Existence of Minimal and Maximal Alternatives)

Let X be a nonempty set and let $\succsim$ be a preference relation on X . At least one member of X is minimal with respect to $\succsim$ in X and at least one member is maximal.

Proposition 1.1 (Representing a preference relation by a utility function)

Every preference relation on a finite set can be represented by a utility function.

When can a preference relation be represented by a utility function?

- What happens when the set X is infinite?
- For many examples, we can find a utility function representation.
- However, not all preference relations on infinite sets can be represented by utility functions.

Example 1.4 (Lexicographic preference relation)

The (lexicographic) preference relation $\succsim$ on $\{(x_1, x_2) : x_1, x_2 \in [0, 1]\}$ defined by $(x_1, x_2) \succ (y_1, y_2)$ if and only if either (i) $x_1 > y_1$ or (ii) $x_1 = y_1$ and $x_2 > y_2$ **is not represented by any utility function**

- We need a more technical condition on preference relation over infinite sets in order to be able ensure the existence of a utility function representation.

- Take the choice set X to be a subset of $\mathbb{R}_+^n$
- The ball around $a \in X$ with radius $r > 0$, denoted $B(a, r)$, is the set of all points in X that are distanced less than r from a .

Definition 1.5 (Continuity)

A preference relation $\succsim$ on X is **continuous** if whenever $a \succ b$, there are balls B_a and B_b around a and b respectively, such that for all $x \in B_a$ and $y \in B_b$ we have that $x \succ y$.

- We are now ready to state a very general theorem about utility function representations, stated by Gerard Debreu (see Debreu (1954,1960) if interested).
- The most general version of the theorem, and its proof, requires more mathematical background¹ than what you are supposed to have for this class. However, I will state, but not prove, a simpler version of the theorem.
- For our purposes, it is sufficient to understand the content of the version I state below.

Proposition 1.2 (Debreu)

Let $\succsim$ be a continuous preference relation on X , which is a convex subset of $\mathbb{R}_+^n$. Then $\succsim$ has a continuous utility representation.

¹The concept of a dense set, which most likely you would see in a Real Analysis class.

- This is enough background material that we will need to begin our study of preferences under uncertainty.
- The goal in this review was to review the concept of a preference relation and the possibility of representing them via a utility function.